

SUGAR

Hypo- & Hyperglycemia



Plan for this session

Case 1 review: 10:30 am

Case 2 review: 10:55 am

Case 3 review: 11:20 am

Case 4 review: 11:40 am

Patient 1:

The parents of a four-year-old boy brought him to the Emergency Department because he was lethargic, complaining of a stomachache and urinating a lot. He was scheduled to see his pediatrician in the morning because he was irritable and pulling on his left earlobe, but when his symptoms changed, the pediatrician recommended taking him to the hospital.

The child's vital signs are pulse 140/min (weak); temperature 102.0°F; respiratory rate 32/min; blood pressure 80/50 mm Hg. The child's abdomen is tender, and there is an effusion from the left middle ear. Skin turgor is low on both the calf and forearm. Capillary refill time on the sternum is 3.5 seconds. Mucous membranes of the mouth appear dry. A finger stick blood glucose returns at 480 mg/dl, and ketone bodies in urine register as 3+. A blood sample is drawn for CBC, BMP, and arterial blood gases.

Recap: Patient 1, Question 1

Based on the HPI & PE, which of the following is on your differential diagnosis for this patient?

- A. Appendicitis
- B. Asthma
- C. Diabetic ketoacidosis
- D. Otitis media
- E. Pneumonia
- F. Urinary tract infection

Evidence for answer “C”: Weak pulse, tachycardia, hypotension, low skin turgor, delayed capillary refill, dry mucous membranes and polyuria are all consistent with dehydration. Tachypnea with deep breaths, tender abdomen and elevated ketone bodies are consistent with ketoacidosis. Elevated blood glucose confirms DKA.

Evidence for answer “E”: Elevated temperature and effusion from left ear are consistent with Otitis media. The ear infection likely triggered the DKA.

Recap: Patient 1, Question 2

Indicate laboratory evidence for hyperglycemia

Analyte	Patient	Reference Range
Hematocrit, %	52	41-53 (m)
Hemoglobin, g/dl	18.4	13.5-17.5 (m)
RBC count, million/ μ l	6.2	4.3-5.9 (m)
MCV, fl	84	80-100
MCH, pg/cell	29.7	25.4-34.6
MCHC, g/dl	35.4	32-36
RDW, %	14.4	11.5-14.5
WBC, per μ l	25,000	4,500-11,000
Platelets, per μ l	386,000	150,000-400,000

Sodium is decreased because elevated glucose increases the extracellular water space. The sodium concentration can be corrected by adding 2 mEq/l for each 100 mg/dl of glucose.

Analyte	Patient	Reference Range
Glucose, mg/dl	460	70-99 (fasting)
Na ⁺ , mEq/l	132	136-145
K ⁺ , mEq/l	4.9	3.5-5.0
Cl ⁻ , mEq/l	105	95-105
HCO ₃ ⁻ , mEq/l	10	22-28
BUN, mg/dl	36	7-18
Creatinine, mg/dl	1.8	0.6-1.2

Analyte	Patient	Reference Range
pH	7.15	7.35-7.45
P _a O ₂ , mm Hg	105	75-105
P _a CO ₂ , mm Hg	22	33-45

Recap: Patient 1, Question 3

Which metabolic process(es) contribute to hyperglycemia in this patient?

- A. Excessive gluconeogenesis
- B. Excessive glycogenolysis
- C. Excessive lipolysis and fatty acid β -oxidation
- D. Insufficient glycolysis activation
- E. Reduced oxidative phosphorylation

Insulinopenia causes excessive glucose production via loss of inhibition of gluconeogenesis (answer “A”) and glycogenolysis (answer “B”). Loss of inhibition of lipolysis results in production of alternative fuels from fatty acid β -oxidation (answer “C”) and ketone body oxidation (answer “D”) that reduce glucose consumption. The rate of oxidative phosphorylation (answer “E”) is determined by ATP consumption. There is no reason to suspect that ATP consumption is altered in this patient.

Recap: Patient 1, Activity 3

Indicate laboratory evidence for dehydration

Analyte	Patient	Reference Range
Hematocrit, %	52	41-53 (m)
Hemoglobin, g/dl	18.4	13.5-17.5 (m)
RBC count, million/ μ l	6.2	4.3-5.9 (m)
MCV, fl	84	80-100
MCH, pg/cell	29.7	25.4-34.6
MCHC, g/dl	35.4	32-36
RDW, %	14.4	11.5-14.5
WBC, per μ l	25,000	4,500-11,000
Platelets, per μ l	386,000	150,000-400,000

Analyte	Patient	Reference Range
Glucose, mg/dl	460	70-99 (fasting)
Na ⁺ , mEq/l	132	136-145
K ⁺ , mEq/l	4.9	3.5-5.0
Cl ⁻ , mEq/l	105	95-105
HCO ₃ ⁻ , mEq/l	10	22-28
BUN, mg/dl	36	7-18
Creatinine, mg/dl	1.8	0.6-1.2

Analyte	Patient	Reference Range
pH	7.15	7.35-7.45
P _a O ₂ , mm Hg	105	75-105
P _a CO ₂ , mm Hg	22	33-45

*Hemoglobin, RBCs, BUN & creatinine
get concentrated due to dehydration*

Recap: Patient 1, Question 4

Explain how hyperglycemia causes dehydration.

- A. Aldosterone inhibition
- B. Aquaporin antagonism
- C. Endothelial leakage
- D. Glucose diuresis

Glucose is a hygroscopic osmolyte. At blood glucose concentrations in excess of 180 mg/dl, glucose is inefficiently filtered by kidneys and therefore passes into urine. Excessive glucose in urine traps excess water, leading to diuresis and dehydration. The same is true for ketone bodies, which are inefficiently filtered by kidneys at concentrations in excess of 5 mM. Excess glucose in urine causes polyuria, which is characteristic of DKA (and HHS).

Recap: Patient 1, Activity 4

Indicate laboratory evidence for acidosis

Analyte	Patient	Reference Range
Hematocrit, %	52	41-53 (m)
Hemoglobin, g/dl	18.4	13.5-17.5 (m)
RBC count, million/ μ l	6.2	4.3-5.9 (m)
MCV, fl	84	80-100
MCH, pg/cell	29.7	25.4-34.6
MCHC, g/dl	35.4	32-36
RDW, %	14.4	11.5-14.5
WBC, per μ l	25,000	4,500-11,000
Platelets, per μ l	386,000	150,000-400,000

Metabolic acidosis consumes bicarbonate, which lowers its concentration in serum. Since bicarbonate is an important physiological buffer, its reduced concentration decreases blood pH. Bicarbonate is also in equilibrium with CO_2 . As the concentration of bicarbonate decreases, the $P_a\text{CO}_2$ also decreases.

Analyte	Patient	Reference Range
Glucose, mg/dl	460	70-99 (fasting)
Na^+ , mEq/l	132	136-145
K^+ , mEq/l	4.9	3.5-5.0
Cl^- , mEq/l	105	95-105
HCO_3^- , mEq/l	10	22-28
BUN, mg/dl	36	7-18
Creatinine, mg/dl	1.8	0.6-1.2

Analyte	Patient	Reference Range
pH	7.15	7.35-7.45
$P_a\text{O}_2$, mm Hg	105	75-105
$P_a\text{CO}_2$, mm Hg	22	33-45

Recap: Patient 1, Question 5

Calculate the anion gap and predict $P_a\text{CO}_2$ from Winter's formula

1 7

Anion gap

2 3

$P_a\text{CO}_2$



Anion gap = $[\text{Na}^+] - [\text{Cl}^-] - [\text{HCO}_3^-] = 132 \text{ mEq/l} - 105 \text{ mEq/l} - 10 \text{ mEq/l} = 17 \text{ mEq/l}$, which is **HIGH** (i.e., > 12)

Winter's formula: $P_a\text{CO}_2$ (estimated) = $1.5 \times [\text{HCO}_3^-] + 8 = 1.5 \times 10 \text{ mEq/l} + 8 = 23 \text{ mm Hg } (\pm 2)$

Since $P_a\text{CO}_2$ (estimated) $\sim P_a\text{CO}_2$ (measured), **respiratory compensation is appropriate**

Recap: Patient 1, Question 6

Based on the laboratory data, which of the following best describes the patient's acidosis?

- A. High anion gap metabolic acidosis
- B. Hyperchloremic, non-anion gap acidosis
- C. Metabolic acidosis with respiratory alkalosis
- D. Non-gap metabolic acidosis
- E. Respiratory acidosis

The laboratory data indicate a high anion gap acidosis with appropriate respiratory compensation (i.e., no respiratory acidosis or alkalosis).

Recap: Patient 1, Question 7

Which of the following is responsible for depressing the pH of arterial blood?

- A. Decreased ammonia excretion
- B. Deep and rapid respiration
- C. Excessive release of fatty acids from adipose tissue
- D. Increased H^+ production from fatty acid β -oxidation
- E. Production of ketone bodies in excess of consumption

Ketone bodies enter the blood (through a transporter) as acids and dissociate into an anion and a proton. When ketone bodies leave the blood and enter cells, they leave as acids (i.e., 'they take the proton with them'). When the rate of ketone body production = rate of ketone body oxidation, there is no acidosis. Acidosis is observed when the rate of ketone body production > the rate of ketone body oxidation.

Recap: Patient 1, Question 8

What's the greatest acute threat to our patient's life?

A. Acidosis

B. Dehydration

C. Hyperglycemia/hyperosmolality

D. Infection/sepsis

E. Ketonuria

Although all of these problems are significant and severe, a high concentration of protons (i.e., a low pH) is acutely neurotoxic and therefore must be treated first.

Recap: Patient 1, Question 9

What's the best way to reduce acidosis in our patient?

- A. Acetazolamide
- B. Dextrose
- C. Half-normal saline
- D. Insulin**
- E. Sodium bicarbonate

Insulin inhibits lipolysis, thus diminishing the release of fatty acids from the adipose tissue. In turn, this reduces fatty acid oxidation in the liver, which then reduces the rate of ketone body synthesis. Isotonic saline is also indicated to treat dehydration. Sodium bicarbonate is only indicated in extreme acidosis ($\text{pH} < 7.0$).

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Patient 2:

The attending is called to another exam room to see a six-month-old patient who was brought to the hospital via ambulance. The patient's mother explains that she called the ambulance because the baby had a seizure. She had a second seizure while the ambulance was en route.

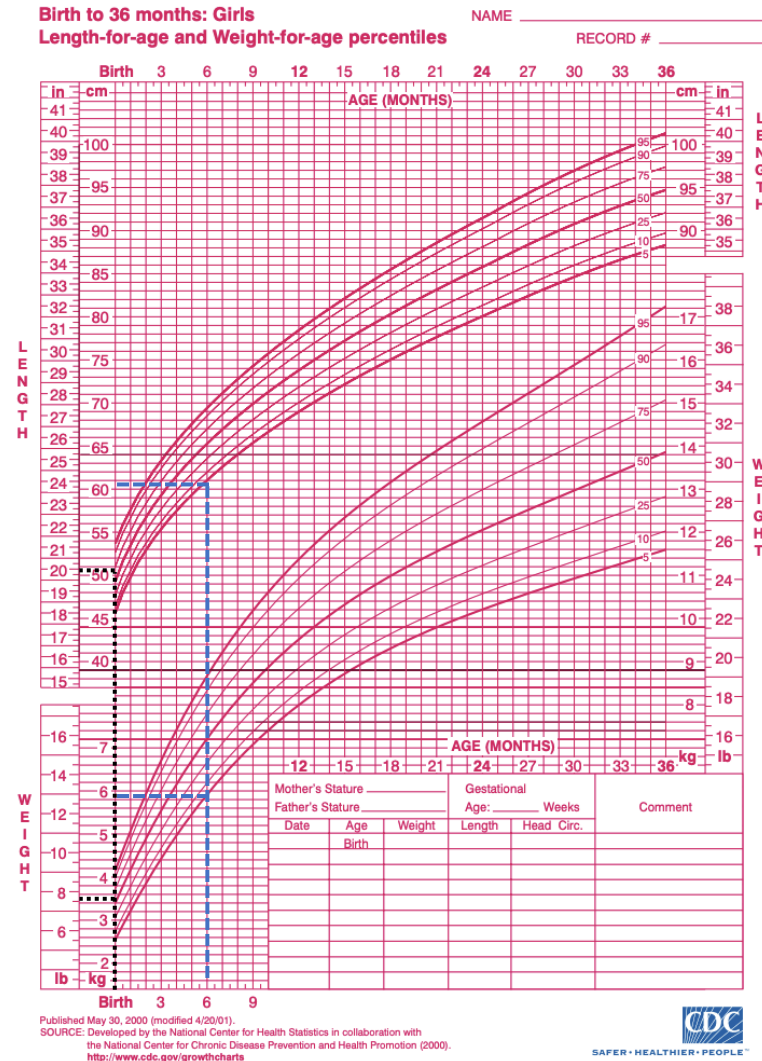
Vital signs: Weight 5.9 kg; Height 61 cm; Temperature 97.0 °F, Respiration Rate 32/min; Pulse 140/min. Physical exam: Child is lethargic, but not cyanotic. Liver is palpable 4 cm below the costal margin. Point of care meter records blood glucose at 18 mg/dl. A Ketostix test registers trace ketone bodies. Blood samples are drawn for CBC, BMP, lipid panel, ammonia, ketone panel, and blood gasses. The baby is given oxygen and started on IV glucose.

When asked about her last feeding, the mother responds that she breast fed her daughter eight hours ago. Five days ago, she began having solid food (once a day, Beechnut, Stage 1, Pears or Apples). The mother has been concerned because her daughter has only grown five inches and gained five pounds since birth.

Analyte	Patient	Reference Range
Hematocrit, %	43	31-55 (f)
Hemoglobin, g/dl	12.5	9.5-13.5 (f)
RBC count, million/ μ l	4.2	3.0-5.4 (f)
MCV, fl	102	85-123
MCH, pg/cell	29.8	28-40
MCHC, g/dl	29.1	29-37
RDW, %	14.5	11.5-16.0
WBC, per μ l	13,200	6,000-17,500
Platelets, per μ l	260,000	150,000-450,000
Glucose, mg/dl	8	70-99 (fasting)
Na ⁺ , mEq/l	134	136-145
K ⁺ , mEq/l	4.5	3.5-5.0
Cl ⁻ , mEq/l	98	95-105
HCO ₃ ⁻ , mEq/l	9.2	20-28
BUN, mg/dl	4.0	7-18
Creatinine, mg/dl	0.45	0.6-1.2

Analyte	Patient	
Ammonia	Within normal range	
Acetone	Slightly elevated	
Cyclohexanone	Within normal range	
Diisobutyl Ketone	Within normal range	
Methyl Ethyl Ketone	Within normal range	
Methyl Isoamyl Ketone	Within normal range	
Methyl Isobutyl Ketone	Within normal range	
Methyl n-Amyl Ketone	Within normal range	
Methyl n-Butyl Ketone	Within normal range	
Analyte	Patient	Reference Range
Total cholesterol, mg/dl	190	Acceptable <110; High >130
Triglycerides, mg/dl	150	Desirable <75; High >100 mg/dL
HDL cholesterol, mg/dl	38	Acceptable >45
LDL-cholesterol, mg/dl	122	Optimal <100; very High >190
Analyte	Patient	Reference Range
pH	7.28	7.38-7.42
PaO ₂ , mm Hg	95	75-100
PaCO ₂ , mm Hg	22	38-42

Recap: Patient 2, Questions 10 & 11



Given this child's length and weight, what are her growth percentiles?

~ 5th percentile

What were her growth percentiles at birth?

~ 50th percentile

Failure to thrive is characteristic of metabolic defects.

Recap: Patient 2, Question 12

Calculate the anion gap and predict $P_a\text{CO}_2$ from Winter's formula

2 7

Anion gap

2 2

$P_a\text{CO}_2$



Anion gap = $[\text{Na}^+] - [\text{Cl}^-] - [\text{HCO}_3^-] = 134 \text{ mEq/l} - 98 \text{ mEq/l} - 9.2 \text{ mEq/l} = 26.8 \text{ mEq/l}$, which is **HIGH** (i.e., > 10)

Winter's formula: $P_a\text{CO}_2$ (estimated) = $1.5 \times [\text{HCO}_3^-] + 8 = 1.5 \times 9.2 \text{ mEq/l} + 8 = 21.8 \text{ mm Hg } (\pm 2)$

Since $P_a\text{CO}_2$ (estimated) $\sim P_a\text{CO}_2$ (measured), **respiratory compensation is appropriate**

Recap: Patient 2, Question 13

Based on the laboratory data, which of the following best describes the patient's acidosis?

- A. High anion gap metabolic acidosis
- B. Hyperchloremic, non-anion gap acidosis
- C. Metabolic acidosis with respiratory alkalosis
- D. Non-gap metabolic acidosis
- E. Respiratory acidosis

The laboratory indicate a high anion gap acidosis with appropriate respiratory compensation (i.e., no respiratory acidosis or alkalosis).

Recap: Patient 2, Question 14

An elevated concentration of which of the following is likely to be responsible for this patient's acidosis?

- A. Acetic acid
- B. Acetoacetic acid
- C. Amino acids
- D. Fatty acids
- E. Formic acid
- F. Lactic acid

The trace of ketone bodies in the blood by Ketostix, and the lightly elevated acetone in the blood suggest that the concentration of ketone bodies is close to normal. Hence, acetoacetic acid is unlikely to be the culprit. Amino acids and fatty acids are not known to cause a metabolic acidosis (the concentration of fatty acids rarely goes beyond about an apparent 2 mM, and most of these fatty acids are bound to albumin, not dissolved in water). A 6-month-old infant is unlikely to have ingested ethanol (acetic acid) or methanol (formic acid). That leaves lactic acid.

Recap: Patient 2, Question 15

What sort of metabolic deficiency is most consistent with this patient's symptoms?

- A. Fatty acid oxidation
- B. Fructose metabolism
- C. Gluconeogenesis
- D. Glycogenolysis
- E. Ketone body synthesis
- F. Oxidative phosphorylation

Since this patient is severely hypoglycemic in the fasting state, they must have a problem mobilizing glucose from glycogen. However, glycogen storage defects generally only cause transient hypoglycemia, so the presentation suggests that gluconeogenesis must also be deficient. For a defect in fatty acid oxidation, the symptoms developed a bit too soon. For fructose metabolism, a problem would likely have manifested sooner than 5 days after introducing fruit. For ketone body synthesis, symptoms developed far too soon. Problems with oxidative phosphorylation should be evident postprandially, not during fasting.

Recap: Patient 2, Question 16

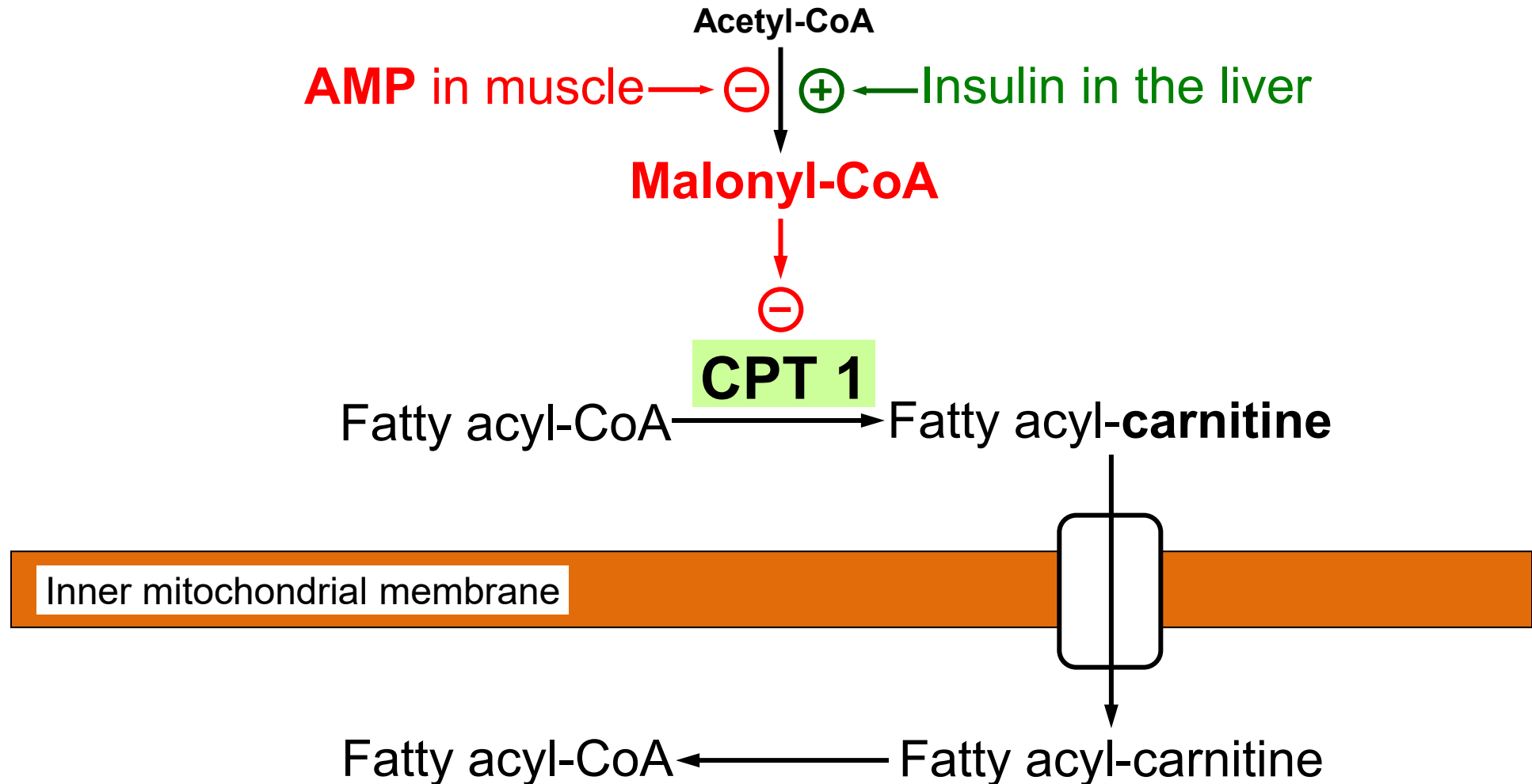
What gene mutation is the most common cause for this type of metabolic defect?

- A. Aldolase B
- B. Galactose-1-phosphate uridylyltransferase
- C. Glucose-6-phosphatase
- D. Glucose-6-phosphate dehydrogenase
- E. Glucokinase

Glucose 6-phosphatase is the only enzyme listed that plays a role in glycogenolysis and gluconeogenesis. It converts glucose 6-phosphate to glucose and is present only in the liver and kidneys. G6Pase deficiency (von Gierke disease) is the most common disorder of glycogenolysis and gluconeogenesis. Although glucose-6-phosphate dehydrogenase (G6PD) sounds similar, it is a completely different enzyme in a completely different pathway (pentose phosphate) expressed in a completely different tissue (RBCs) that causes a completely different pathology (hemolysis).

Why is this patient non-ketotic?

von Gierke disease is non-ketotic because high acetate drives synthesis of malonyl-CoA



Recap: Patient 2, Question 17

How can we confirm this hypothesis?

- A. Assay of blood sample
- B. Assay of liver biopsy
- C. Assay of muscle biopsy
- D. Cell-free DNA screening
- E. Karyotype
- F. Whole exome sequencing

Deficiency in glucose-6-phosphatase can be measured at the level of enzyme activity with a tissue sample from an organ that expresses the enzyme (liver or kidney), or by sequencing the gene (which is present in every nucleated cell). Note that although muscle synthesizes and degrades glycogen, it does NOT express glucose-6-phosphatase because it does not release glucose to circulation (muscle uses all the glycogen it stores).

Recap: Patient 2, Question 18

What accounts for triglyceridemia in this patient?

- A. Hyperinsulinemia stimulates VLDL production in the liver
- B. Hypoinsulinemia blocks lipoprotein lipase activity on chylomicrons
- C. Increased fatty acid synthesis from trapped hepatic glucose
- D. Increased synthesis of glycerol-3-phosphate

Trapped hepatic glucose stimulates flux in glycolysis, which raises the concentration of all glycolytic intermediates, including glycerol-3-phosphate (G3P). Both fatty acid and glycerol synthesis from trapped hepatic glucose is increased. Until this patient is 2 years old, she will receive glucose overnight continuously via a nasogastric infusion line. Starting at 2 years of age, she will receive uncooked corn starch at regular intervals. The hypertriglyceridemia should resolve by the time the patient transitions to oral feeding.

Recap: Patient 1, Question 19

The attending physician is called back to the first patient's room because of emerging hypokalemia and hypophosphatemia following insulin treatment.

What acute risk(s) does hypophosphatemia increase in patients?

A. Aortic dissection

B. Cardiac arrest

C. Hypoxia

D. Pulmonary edema

E. Respiratory failure

Hypophosphatemia compromises energy metabolism, which is critical for conduction in the nervous system and for muscle contraction. It also reduces oxygen perfusion and respiration in the CNS.

Recap: Patient 1, Question 20

How does hypophosphatemia compromise energy metabolism?

- A. Blocks ATP synthesis
- B. Blocks DNA synthesis
- C. Blocks fatty acid oxidation
- D. Blocks glycogenolysis
- E. Blocks glycolysis

Phosphate is a substrate for ATP synthase. Without phosphate, ATP cannot be made. Phosphate is also a substrate for glyceraldehyde-3-phosphate dehydrogenase, which is necessary for glycolysis. Phosphate is a substrate for glycogen phosphorylase, which liberates glucose from glycogen, but this effect is far less dangerous to the patient than the first two. Although DNA synthesis and fatty acid oxidation would also be inhibited, these effects are secondary to inhibition of ATP synthesis.

Recap: Patient 1, Question 21

Why does hypophosphatemia cause hypoxia without hypoxemia?

- A. Decreased activity of glucose-6-phosphate dehydrogenase
- B. Decreased concentration of 2,3-bisphosphoglycerate
- C. Decreased concentration of NADH
- D. Decreased pH of blood

Hypophosphatemia reduces the activity of glyceraldehyde 3-phosphate dehydrogenase and hence reduces the concentrations of 1,3-bisphosphoglycerate and 2,3-bisphosphoglycerate (2,3-BPG), thereby increasing the affinity of hemoglobin for O₂.

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Case 3 review: 11:20 am

Case 4 review: 11:40 am

Patient 3:

A 7-year-old patient presented to the ED with polyuria (3-4 times per hour for the last 3 days) and lethargy. She had recently moved to a new house and was having a hard time fitting in at her new school. Her weight was in the 5th-10th percentile. Pulse: 88/min; RR: 18/min; BP: 100/60 mm Hg; SpO₂ (room air): 100%. Initial lab tests showed: glucose: 1,167 mg/dL, osmolality: 350 mOsm/kg, venous pH 7.34, sodium 123 mEq/L. Urine dipstick: strongly positive for glucose, mildly positive for ketone bodies.

Recap: Patient 3, Question 22

Based on the laboratory data, what is the most likely diagnosis for this patient's condition?

- A. Bulimia nervosa
- B. Diabetic ketoacidosis (DKA)
- C. Hyperosmolar, hyperglycemic state (HHS)
- D. Urinary tract infection (UTI)

The patient has very high blood sugar, but near normal pH and minimal ketone bodies, suggesting HHS. Polyuria is likely related to high blood sugar. Growth retardation may be a consequence of either physiological or psychological factors.

Recap: Patient 3, Question 23

How should this patient be treated?

A. Antibiotics

B. Fluids

C. Insulin

D. Metformin

The patient's primary issues are dehydration and elevated blood glucose. Fluids are necessary to relieve dehydration, and insulin will effectively resolve elevated blood sugar.

Recap: Patient 3, Question 24

What is the likely pathophysiology of this patient's condition?

- A. Cushing disease
- B. Maturity-onset diabetes of the young (MODY)
- C. Medium-chain acyl-CoA dehydrogenase (MCAD) deficiency
- D. New-onset type 1 diabetes
- E. von Gierke disease

The patient has adequate insulin to prevent ketoacidosis, but insufficient insulin to block endogenous glucose production. Given her age, new-onset type 1 diabetes is the most likely etiology, but MODY is a possibility. The patient's physical presentation is inconsistent with Cushing disease. MCAD deficiency and von Gierke disease cause hypoglycemia.

Recap: Patient 3, Question 25

What is the most important next step for managing this patient?

- A. Antibiotics
- B. Autoantibody testing
- C. Corticosteroids
- D. Counseling
- E. Genetic testing

Autoantibody testing is necessary to confirm a diagnosis of new-onset type 1 diabetes. If this test returns negative, genetic testing for MODY would be the next step. Regardless of diagnosis, the patient and her family will benefit from counseling to help them manage her condition. Antibiotics are probably unnecessary for this patient. Corticosteroids are contraindicated.

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Case 3 review: 11:20 am

Case 4 review: 11:40 am

Recap: Patient 4 (you!)

If your pre-TBL blood glucose was 195 mg/dl and you ate one chocolate glazed and one glazed munchkin during TBL, then had a chicken burrito with white rice, black beans, cheese, guacamole and pico de gallo at Chipotle for lunch, how much insulin would you need to inject to correct your blood glucose?

For pre-TBL hyperglycemia, you need $(195\text{mg/dl} - 150\text{mg/dl})/30\text{mg}/(\text{dl} \cdot \text{U}) = 1.5 \text{ units}$.

To correct for the munchkins, you need $(8\text{g} + 7\text{g})/14\text{g}/\text{U} = 1.1 \text{ units}$

(carbohydrate content of munchkins from www.dunkindonuts.com/en/menu/donuts/product-munchkins-id1001204).

To correct for the burrito, you need $126\text{g}/14\text{g}/\text{U} = 9 \text{ units}$

(carbohydrate content of the burrito from <https://www.chipotle.com/nutrition-calculator/burrito>).

*Therefore, you need to inject **11.6 units of insulin** to correct your blood glucose.*